

- $\Sigma = \Sigma(\omega)$ o.d. $\Sigma(\omega) = n^2(\omega)$ (omni kotičnik)
- $\Sigma(\omega) \in \mathbb{C}$: $\text{Re} \Sigma(\omega)$ določa odboj in lom
- $\text{Im} \Sigma(\omega)$ predstavlja absorpcijo v snovi

• Kramers-Kronigove relacije

$$\begin{aligned} \text{Re} \Sigma(\omega) &= 1 + \frac{2}{\pi} \mathcal{P} \int_0^\infty \frac{\omega' \text{Im} \Sigma(\omega')}{\omega'^2 - \omega^2} d\omega' \\ \text{Im} \Sigma(\omega) &= -\frac{2\omega}{\pi} \mathcal{P} \int_0^\infty \frac{\text{Re} \Sigma(\omega') - 1}{\omega'^2 - \omega^2} d\omega' \end{aligned}$$

$$\mathcal{P} \int_{-\infty}^{\infty} \frac{g(\omega')}{\omega' - \omega} d\omega' = \lim_{\epsilon \rightarrow 0} \left[\int_{-\infty}^{\omega - \epsilon} \frac{g(\omega')}{\omega' - \omega} d\omega' + \int_{\omega + \epsilon}^{\infty} \frac{g(\omega')}{\omega' - \omega} d\omega' \right]$$

• izogiba se polu pri $\omega = \omega'$

• nepopolni v krajji, preko integrala glavne vrednosti, Plancherjevih enačb in Hilbert. prost.

• Modeli $\Sigma(\omega)$

• Gibalna en. za vzrazn. nihaj

$$m \frac{d^2 \vec{r}}{dt^2} = -m \gamma \frac{d\vec{r}}{dt} - m \omega_0^2 \vec{r} + e \vec{E}(t)$$

$\downarrow$ $\downarrow$ $\downarrow$
 disperzija harmonski zun. polje
 kot posledica potencial
 vzbujenosti
 ali sipanja
 v kristalih

Fourier

$$-m \omega^2 \vec{r} = m \gamma i \omega \vec{r}(\omega) - m \omega_0^2 \vec{r}(\omega) + e \vec{E}(\omega) \rightarrow \vec{r}(\omega) = \frac{e}{m} \frac{\vec{E}(\omega)}{\omega_0^2 - \omega^2 - i \gamma \omega}$$

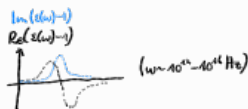
$\vec{r} = \frac{e}{m} \frac{\vec{E}(\omega)}{\omega_0^2 - \omega^2 - i \gamma \omega}$ $\uparrow$
 ud. gost.

$$\Sigma(\omega) = \epsilon(\omega) = 1 + \frac{ne^2}{m} \frac{1}{\omega_0^2 - \omega^2 - i \gamma \omega}$$

- v realni snovi je lahko več vrst harmonskih potencialov (vz. ω_0) in več vrst disperzije (γ)
- obstaja več režimov (približkov):
Debyeov./Lorentsov./plazemski nileksacija

→ Debyeova relaksacija (nizke frekvence, zanemarimo ω_0): $\vec{r}(\omega) = \frac{ne^2}{m} \frac{\vec{E}(\omega)}{\omega_0^2 - \omega^2 - i \gamma \omega}$ ($\omega \ll \omega_0 \approx 10^{14} \text{ Hz}$)

→ Lorentsov relaksacija (puna odvisnost, uprabi vs. člene): $\vec{r}(\omega) = \frac{ne^2}{m} \frac{\vec{E}(\omega)}{\omega_0^2 - \omega^2 - i \gamma \omega} \rightarrow \epsilon(\omega) = 1 + \frac{(\epsilon(\infty) - 1) \omega_0^2}{\omega_0^2 - \omega^2 - i \gamma \omega}$



→ Plazemski relaksacija (visoka frekvence, zanemarimo vse razen el. gonjanj): $\vec{r}(\omega) = \frac{ne^2}{m} \frac{\vec{E}(\omega)}{\omega^2} \rightarrow \epsilon(\omega) = 1 - \frac{\omega_p^2}{\omega^2}$ (plazemski frekvenci, $\omega_p^2 = \frac{ne^2}{m \epsilon_0}$) ($\omega > 10^{14} \text{ Hz}$)